Mendelian randomization and multi-omic evidence for regulatory peak–gene links in human immune cells

Short title: Genetic evidence for immune-cell peak–gene links

PLOS Genetics Research Article — working draft

TODO: Supply author names in agreed order, numbered affiliations, and corresponding author name and email. Add author notes only when applicable. Do not infer authorship or affiliations from the project files.

Abstract

Placeholder. The abstract will be written after the Results narrative and figure selection have been reviewed.

TODO: Summarize the objective, three immune cell types, six-method comparison, reference-based evaluation, and functional and disease-associated examples. Separate the number of tested pairs from the number of significant links; avoid a general claim that MR outperforms marginal association.

Author Summary

TODO: Write a 150–200-word, nontechnical Author Summary in the first person after the scientific narrative is approved. This is a placeholder, not a completed summary. The unstructured Abstract must be no more than 300 words.

Introduction

Preliminary paragraph outline.

1. **Genetic regulation of expression.** Introduce cis-regulatory variation and the use of expression QTLs to connect genetic variation with molecular phenotypes.

2. **Accessible chromatin and regulatory elements.** Explain how chromatin-accessibility peaks nominate regulatory regions, while accessibility alone does not specify the regulated gene.

3. **The peak–gene linking problem.** Introduce individual-level pseudobulk and multiome metacell correlations, including the distinction between within-cell-type and across-cell-type associations.

4. **Genetic instruments as complementary evidence.** Motivate MR using caQTLs as exposure associations and eQTLs as outcome associations. Introduce MR-Egger and summary-data-based MR as related methodological foundations [1, 2]; their use does not by itself establish the validity of every inferred regulatory link.

5. **Study objectives.** Compare four MR-based tests and two marginal correlation methods in naive B, CD4 T, and CD8 T cells; evaluate regulatory reference support; and explore functional annotations and disease-associated loci without presupposing a universally superior method.

TODO: Add the CIMA study citation, accessibility and peak–gene linking background references, and the external PBMC multiome source. Develop these paragraphs after agreeing on the central scientific message.

Results

A common analysis set enables comparison of six linking methods

We used cell-type-specific CIMA caQTL and eQTL summary statistics to investigate accessibility-to-expression links in naive B cells (Bn), naive CD4 T cells (CD4), and naive CD8 T cells (CD8). The six methods were GLS-IVW, GLS-MR Egger, SMR, an equal-weight Cauchy combination of these three MR tests (CCT), WNN metacell Spearman correlation, and donor-level pseudobulk Spearman correlation.

After requiring shared variants and feature availability in both marginal datasets, retaining complete base-method p-values, and applying the IVW residual-variance filter, the common analysis sets comprised 40,087, 85,527, and 50,931 peak–gene pairs in Bn, CD4, and CD8, respectively (Table 1). These are *tested*

pairs, not counts of significant regulatory links. Throughout the discovery comparison, q denotes a p-value adjusted by Benjamini–Hochberg (BH) separately within each gene and method.

Table 1. Retained peak–gene analysis sets. Counts refer to the final common analysis table. WNN uses the same 80 external PBMC metacells for all three cell types; donor counts refer to the cell-type-specific pseudobulk data.

Cell type	Tested pairs	Genes	Peaks	Donors	Metacells
Bn	40,087	8,149	2,951	335	80
CD4	85,527	10,701	5,338	372	80
CD8	50,931	9,427	3,327	346	80

At $q < 0.1$, WNN identified the largest number of significant links in each cell type (Table 2). CCT identified more links than any individual MR test, whereas pseudobulk identified more links than CCT in each cell type. These counts describe differences in detection and do not establish comparative accuracy.

Table 2. Significant links at within-gene BH-adjusted $q < 0.1$. Each count is a subset of the tested pairs in Table 1; method-specific sets overlap.

Method	Bn	CD4	CD8
GLS-IVW	3,977	9,569	5,722
GLS-MR Egger	3,361	7,265	4,538
SMR	3,534	8,627	5,220
CCT	4,406	10,597	6,451
WNN	18,179	37,181	24,114
Pseudobulk	4,924	17,181	9,618

Shared links can have different MR and marginal effect directions

The methods identified overlapping and method-specific link sets (Figure 1). All six methods were significant for 469 Bn, 1,259 CD4, and 759 CD8 pairs. Among these shared links, the three MR effect estimates agreed in sign for 469/469, 1,258/1,259, and 758/759 pairs, respectively. IVW agreed in sign with *both* marginal correlations for 354/469 (75.5%), 918/1,259 (72.9%), and 564/759 (74.3%) pairs.

Selected examples display concordant signals, weak marginal associations, and opposite association directions (Figure 2). For example, the CD8 ERAP2 example shows negative MR slopes and a negative pseudobulk correlation, but a positive WNN correlation. These patterns motivate examining the different data levels rather than treating the methods as interchangeable measurements.

TODO: Develop biological interpretation of the selected examples and assess whether the current four-row example page should be simplified.

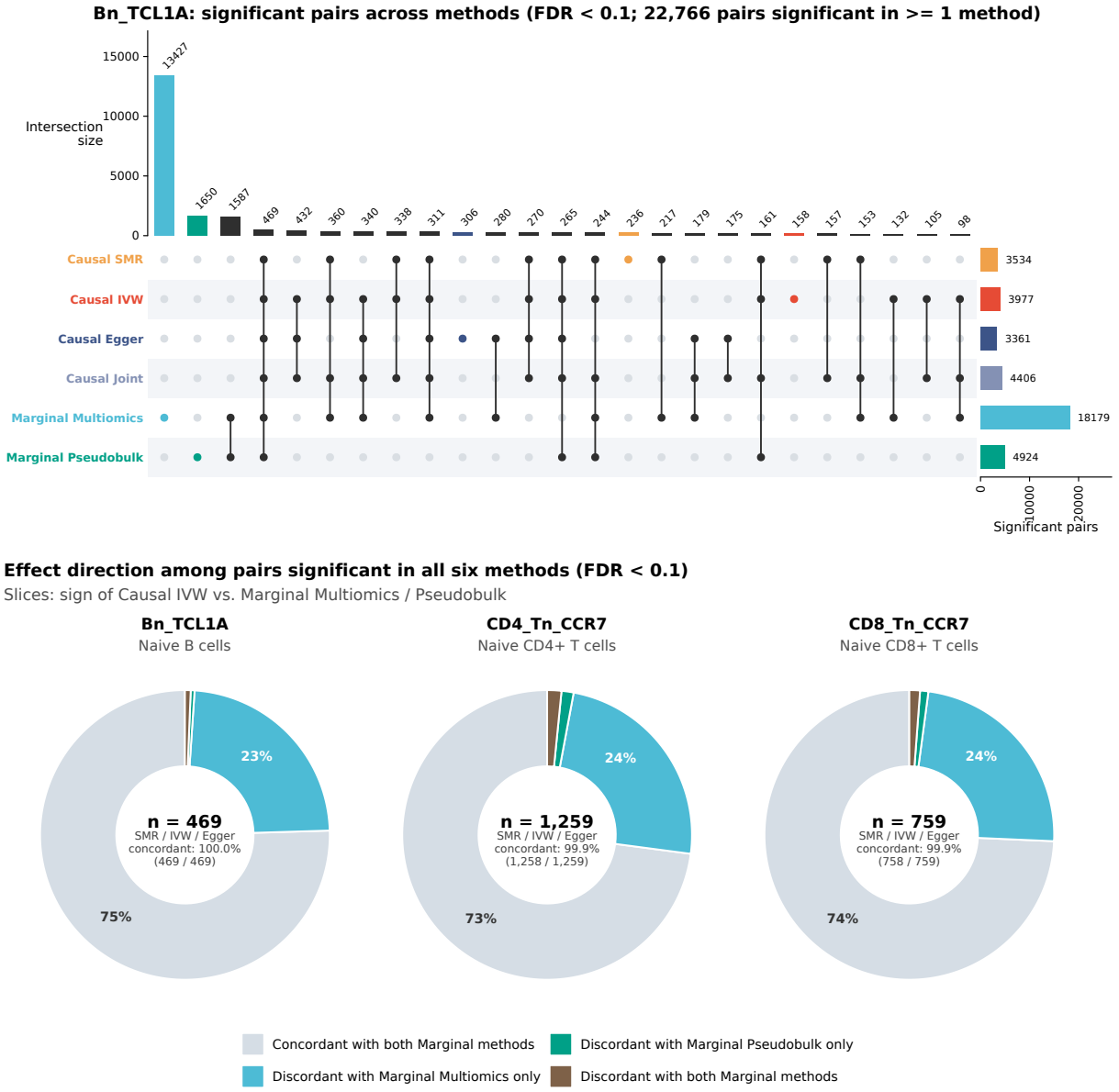


Fig 1. Overlap and direction of significant links. Top: Bn UpSet plot, showing the 25 largest distinct significance combinations at $q < 0.1$. Bottom: effect-direction comparison among links significant in all six methods in each cell type. Slices compare the IVW slope sign with the WNN and pseudobulk correlation signs; central annotations summarize agreement among SMR, IVW, and Egger. Existing source pages 1 and 4 are shown unchanged. The figure's "Causal Joint" label denotes CCT.

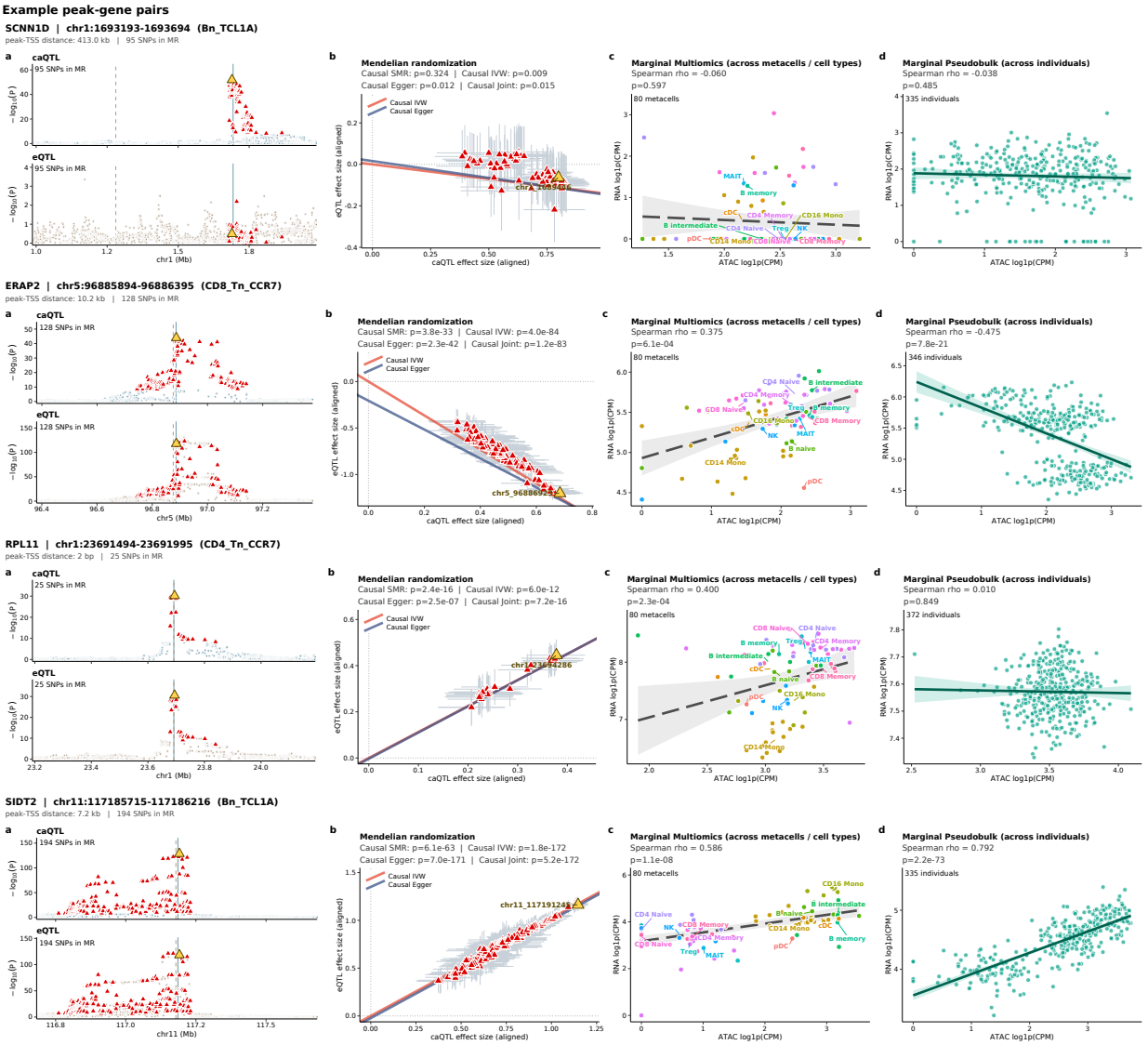


Fig 2. Selected peak–gene examples: SCNN1D in Bn, ERAP2 in CD8, RPL11 in CD4, and SIDT2 in Bn. Columns show QTL association tracks, aligned SNP effects with IVW/Egger fitted lines, WNN metacell measurements, and pseudobulk measurements. The displayed significance summaries are from the existing analysis; marginal correlation summaries are Spearman correlations, while the drawn marginal trend lines are linear fits. In the caQTL tracks, non-instrument variants with $p < 10^{-8}$ are omitted by the original plotting script. Source page 1 is shown unchanged; further examples remain in the copied six-page source PDF.

TODO: revise panel size and caption after example selection is reviewed.

Regulatory reference support depends on the evaluation metric

We evaluated links against EnhancerAtlas, FANTOM5, 4DGenome, GTEx, HiChIP, and PCHiC reference datasets. The current pooled overlap summary reports enrichment above one in every cell-type, method, and reference combination. SMR has the largest pooled fold enrichment in 17 of 18 cell-type/reference combinations; IVW has the largest value for CD4 PCHiC. In Bn, for example, SMR and WNN fold enrichments are 6.18 and 3.56 against FANTOM5, and 4.46 and 3.08 against HiChIP. The full existing summary is supplied as supporting Table S1 (`Fig3_Hyper_table.xlsx`). These are descriptive comparisons of method-specific selected sets, with different numbers of links and contributing genes; they are not paired tests of method superiority.

We also summarized gene-wise ROC-AUC using raw-p-value significance scores (Figure 3). Differences in the stored mean AUCs were smaller and benchmark-dependent. For Bn FANTOM5, mean AUC was 0.813 for SMR and 0.807 for WNN, whereas for CD4 FANTOM5 it was 0.778 and 0.779, respectively. The existing ROC calculations use automatic direction selection, so these summaries should be read as the implemented benchmark rather than a fixed-direction ranking comparison.

TODO: Select a compact reference-support table for the manuscript and develop the interpretation of enrichment versus gene-wise discrimination.

CIMA regulatory networks provide within-cohort corroboration

The cell-type-active CIMA SCENIC+ networks contained 864, 2,520, and 1,364 pairs from the retained Bn, CD4, and CD8 analysis sets, respectively. At $q < 0.05$, WNN had more network-overlapping significant pairs than SMR (513 versus 324 in Bn), but SMR had a larger enrichment odds ratio (9.60 versus 2.30 in Bn). This illustrates the distinction between overlap counts and enrichment relative to the tested background (Supplementary Figure S1). CIMA networks provide corroboration within the same study, rather than independent experimental confirmation.

TODO: Integrate the high-quality eRegulon subset after deciding how much GRN detail belongs in the main text.

Motifs and gene sets characterize functional relevance

We examined motif enrichment in each method's significant peaks using GC- and accessibility-matched backgrounds. Across 746 JASPAR motifs, the IVW foreground yielded 58, 96, and 76 enriched motifs in Bn, CD4, and CD8, respectively (BH-adjusted $p < 0.05$ and fold enrichment > 1). The corresponding WNN

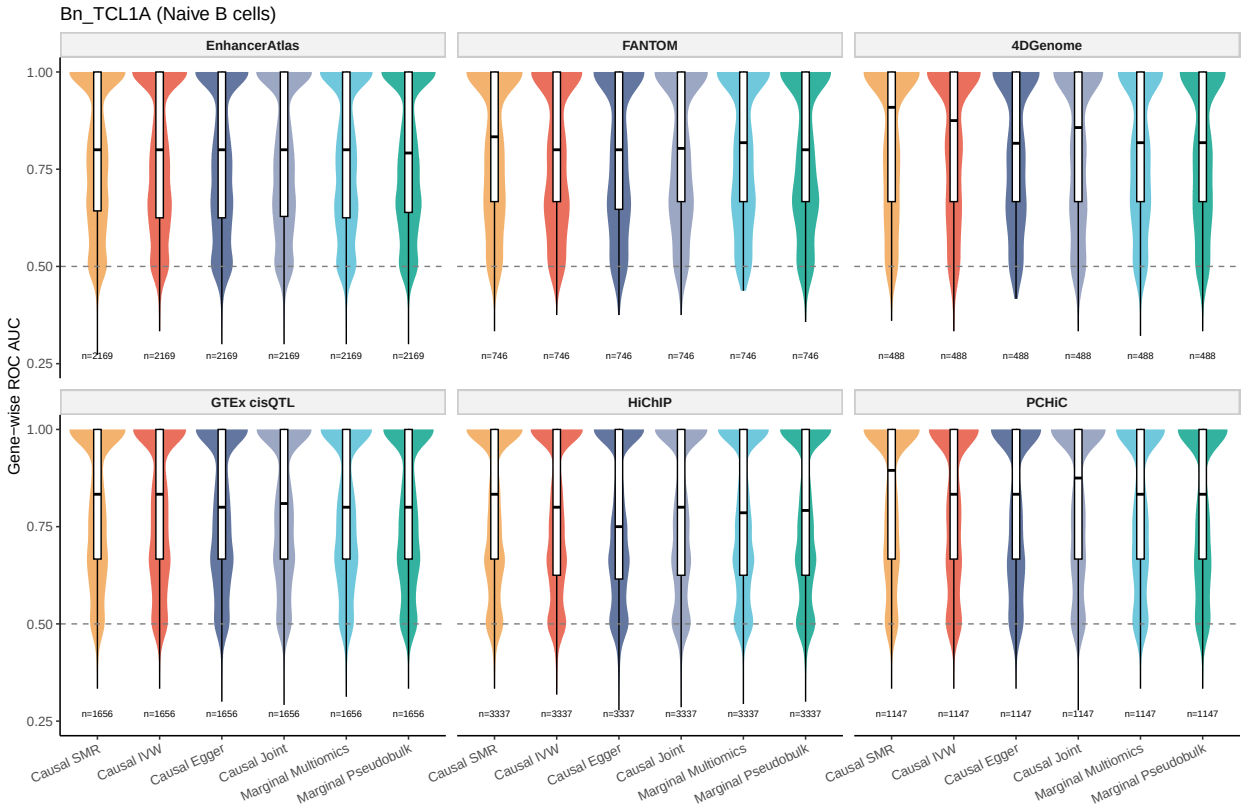


Fig 3. Gene-wise ROC-AUC against six regulatory references in Bn. Violin and box plots summarize genes having both reference-overlapping and other tested peaks; annotations give gene counts. Scores are $-\log_{10}(p)$, and ROC direction was selected automatically in the existing analysis. The dashed line marks AUC 0.5. This is page 1 of the latest available AUC PDF, stored in the older figure directory; CD4 and CD8 are retained as pages 2 and 3 of the copied file.

counts were 92, 123, and 92. These method-specific foregrounds are not restricted to links unique to MR. TF-expression summaries and the existing motif spreadsheet are supplied as supporting material (Table S2).

TODO: Choose representative TF motifs, verify their expression summaries, and develop the biological interpretation; motif counts alone do not establish functional specificity.

For gene-set interpretation, we defined MR-only links as significant in at least one of IVW, Egger, or SMR at $q < 0.1$, with both marginal methods having $q \geq 0.1$. Their target genes numbered 1,581, 2,834, and 1,903, with 4,509 unique genes in the combined set. The existing GO Biological Process GSEA returned no terms at conventional BH-adjusted $p < 0.1$ for individual cell types, and nine for the combined set. Top combined terms include Arp2/3 complex-mediated actin nucleation and purine metabolism (Supplementary Figure S2). A separate 5,000-draw gene-set resampling analysis returned 34, 213, 224, and 499 terms at empirical BH-adjusted $p < 0.1$, respectively. These analyses use different null comparisons and are kept distinct in this draft.

TODO: Decide which GO comparison addresses the intended biological question and select a small set of interpretable terms.

MR-only regions show exploratory disease-heritability enrichment

We used S-LDSC to examine peaks from MR-only links, with the definition above, against nine immune-related GWAS. After liftOver to hg19, the MROnly sets contained 1,271 Bn, 2,433 CD4, and 1,498 CD8 peaks; their union contained 3,777 peaks. Each extended peak annotation and the all-candidate-peak annotation were fitted separately with baseline-LD annotations.

For the combined MROnly annotation, raw enrichment p-values were below 0.1 for Crohn's disease (CD), inflammatory bowel disease (IBD), multiple sclerosis (MS), and type 1 diabetes (T1D), with enrichment estimates of 20.05, 18.04, 50.83, and 19.72, respectively (Figure 4). Their raw p-values ranged from 0.050 to 0.084; none met raw $p < 0.05$. These are exploratory annotation-enrichment findings, without correction across the displayed GWAS comparisons and without a direct test of MROnly versus all peaks.

TODO: Develop the disease interpretation and decide how to report uncertainty alongside the existing figure, which omits standard-error bars.

GWAS-overlapping peaks nominate disease-associated regulatory examples

For locus exploration, a separate, more stringent definition selected links with any IVW, Egger, SMR, or CCT $q < 0.001$ and both marginal $q \geq 0.05$. Requiring a GWAS variant with $p < 5 \times 10^{-8}$ inside the unextended

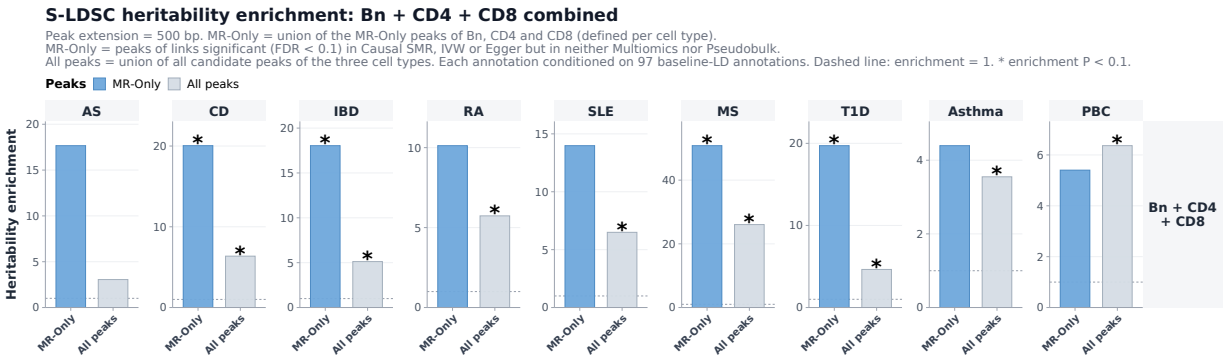


Fig 4. S-LDSC enrichment for the union of the three cell-type MROnly peak sets and the union of all candidate peaks. MROnly uses IVW/Egger/SMR link $q < 0.1$ and both marginal $q \geq 0.1$. Peaks were lifted to hg19 and extended by 500 bp on each side. Each custom annotation was fitted separately with 97 baseline-LD annotations. Asterisks denote raw enrichment $p < 0.1$, not multiplicity-adjusted significance or a test between the two bars. AS: ankylosing spondylitis; CD: Crohn's disease; IBD: inflammatory bowel disease; RA: rheumatoid arthritis; SLE: systemic lupus erythematosus; MS: multiple sclerosis; T1D: type 1 diabetes; PBC: primary biliary cirrhosis, using the source GWAS label. Source page 2 is unchanged; cell-type results are shown in Supplementary Figure S3.

peak identified 27 cell-type-specific peak–gene pairs (7 Bn, 10 CD4, and 10 CD8), covering 21 genes and 14 distinct peaks. The 37 output rows include repeated pairs associated with different variant/trait records.

One example links the hg38 peak chr11:61827740–61828241 to TMEM258 in CD4 cells (Figure 5). SMR and IVW adjusted p-values were 2.39×10^{-7} and 8.21×10^{-6}, while WNN and pseudobulk adjusted p-values were 0.398 and 0.225. The peak contains rs968567, associated with asthma at $p = 3.67 \times 10^{-10}$. This positional overlap nominates a candidate for follow-up; it does not establish colocalization or a mechanism mediating disease risk.

TODO: Review TMEM258 and alternative loci using appropriate biological references before selecting the final disease example.

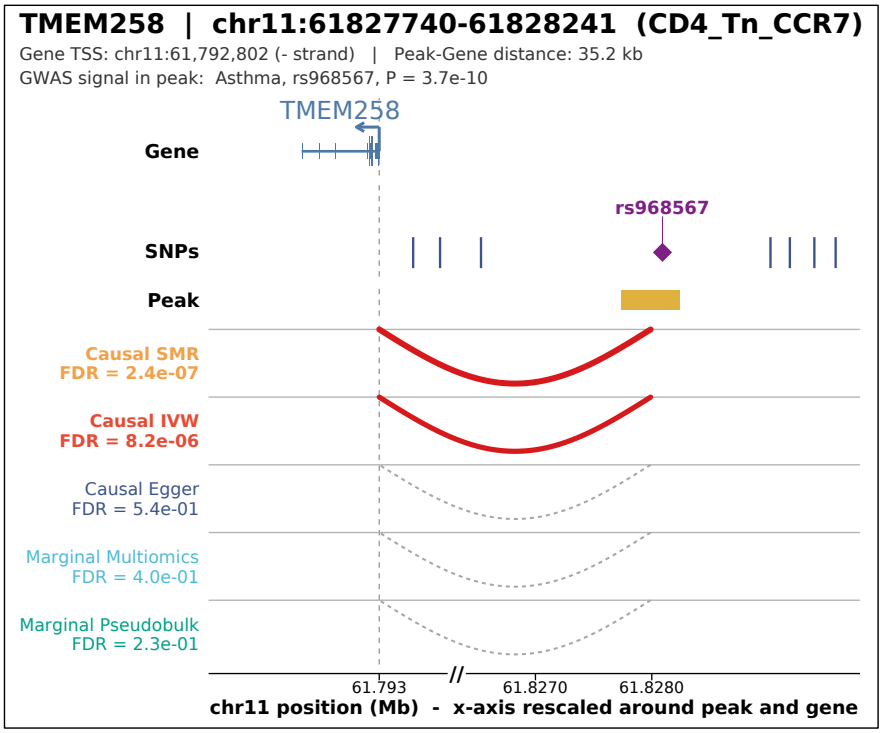


Fig 5. An asthma-associated peak linked to TMEM258 in CD4 cells. Tracks show the gene, retained MR instruments, peak, and five method-specific links; CCT is omitted in the original display. Solid links indicate $q < 0.05$, and dashed links indicate nonsignificance at that display threshold. The purple symbol marks the overlapping GWAS variant rs968567. The x-axis is rescaled around the peak and gene. This is unchanged source page 7 of the 27-page locus PDF; example selection remains provisional.

Discussion

Preliminary outline only.

- **Principal findings.** Discuss differences in discovery yield, overlapping and discordant links, and the distinction between reference enrichment and ranking performance.
- **Complementary biological information.** Consider how genetic evidence, donor-level associations, and across-cell-type metacells address different aspects of regulation. Develop motif, GRN, GO, and disease-locus interpretations after reviewing the selected examples.
- **Scope of inference.** Discuss the assumptions underlying MR, incomplete reference labels, cell-type comparability, and the distinction between positional GWAS overlap and a shared causal mechanism. Present S-LDSC and GO exploration with their stated thresholds and null comparisons.
- **Limitations and future directions.** Outline source-provenance details still to be completed, independent validation, statistical sensitivity analyses if subsequently prioritized, and experimental follow-up.

TODO: Write the Discussion after the Results narrative and figure placement have been reviewed. Do not infer a universal advantage of MR from discovery counts or selected examples.

Materials and Methods

Structured outline; mathematical details and complete source citations will be developed in subsequent revisions.

Data sources

We used CIMA caQTL and eQTL summary statistics (OMIX009786), cell-type-specific pseudobulk matrices, and an external PBMC multiome WNN dataset. The MR LD reference is hg38 1000 Genomes EAS; S-LDSC uses its separate EUR reference.

TODO: Add study citations, donor and QTL sample sizes, published covariates, normalization, ancestry details, and data-access statements.

Candidate peak–gene pairs

Precomputed cell-type-specific candidate tables supplied the pair universe. The documented upstream search includes peaks with centers within 1 Mb of a gene's TSS. Current eligibility requires at least three shared QTL variants and feature availability in both marginal datasets.

TODO: Document the definitive candidate-table generation and coordinate rules.

Instrument selection and LD preparation

The model consumes pair-indexed caQTL files documented as clumped at $r^2 = 0.99$ within 20 Mb. QTL alleles must match exactly; approximately duplicated exposure/outcome slope pairs are removed at tolerance 10^{-3}. Signed PLINK LD is aligned to the retained variants, and exposure effects, outcome effects, and LD signs are oriented together.

TODO: Recover the original clumping command and initial instrument-selection thresholds; give the complete retained-variant rules.

GLS-IVW

IVW is fitted through the origin after whitening by the outcome-effect covariance constructed from outcome SEs and LD. Eigenvalue truncation targets 99.9% variance with a five-direction minimum setting, subject to available positive directions. The implementation uses SNP-count-based degrees of freedom and inflates SEs only for overdispersion.

TODO: Add the GLS formulation, reference for correlated instruments, and the precise truncation and uncertainty conventions.

GLS-MR Egger

Egger uses the same LD-aware transformation and an additional intercept [1]. Final tables retain the slope, SE, and p-value.

TODO: Describe the intercept and assumptions, distinguishing the general MR-Egger reference from this project's GLS implementation.

Summary-data-based Mendelian randomization

The SMR calculation uses the retained SNP with the smallest caQTL p-value, its outcome/exposure effect ratio, and the implemented chi-square test based on p-derived z-scores [2].

TODO: Add the exact test and SE expressions. Clarify that this implementation does not include a HEIDI analysis.

Cauchy combination

CCT combines the SMR, IVW, and Egger p-values with equal weights [3]. It supplies a combined significance measure rather than a combined causal effect estimate.

Pseudobulk and WNN correlations

Pseudobulk Spearman correlations use matched donors and log-transformed CPM. WNN Spearman correlations use 80 external PBMC metacells shared across the three analyses. Neither correlation stage includes covariate residualization.

TODO: Complete the upstream pseudobulk and metacell construction descriptions.

Filtering and multiple testing

Pairs with any missing base-method p-value are removed, then pairs with IVW residual variance greater than 10 are removed. CCT is calculated afterward. BH adjustment is applied separately within each gene and method [4]. Discovery uses $q < 0.1$; GRN uses $q < 0.05$. The two definitions of MR-only links are stated in their Results subsections.

Regulatory references, enrichment, and ROC-AUC

Six reference datasets are mapped to genes and genomic intervals. Gene-level upper-tail hypergeometric tests compare significant peaks with candidate backgrounds; the existing global summary pools eligible gene counts and reports fold enrichment. ROC-AUC uses $-\log_{10}(p)$, genes with both classes, and the existing automatic ROC direction. GTEx ROC labels additionally require overlap with validation multiome peaks.

TODO: Add resource citations, overlap/liftOver details, reference thresholds, and a precise definition of the comparison population.

Motifs, regulatory networks, and GO

Motif analysis uses Signac, JASPAR2020, and backgrounds matched on GC and accessibility; enrichment requires BH-adjusted $p < 0.05$ and fold > 1. TF-expression checks require detection in at least 5% of cells. GRN overlap uses Fisher tests against all retained tested pairs. GO BP GSEA ranks MR-only genes by minimum MR p-value over MR-only links; a separate 5,000-draw analysis samples tested genes scored using all their links and calculates empirical p-values from term-specific available null NES values.

TODO: Add resource/software citations, term-size settings, gene identifiers, and the intended interpretation of each GO null comparison.

Disease heritability and GWAS overlap

S-LDSC [5] evaluates hg19 peak annotations extended by 500 bp per side, each fitted with 97 baseline-LD annotations. The nine GWAS are listed in Figure 4. A separate locus screen requires a genome-wide-significant GWAS variant inside an unextended peak.

TODO: Add each GWAS citation and cohort description, regression resources, uncertainty reporting, and the variant-mapping procedure.

Null-SNP diagnostic

An existing diagnostic LD-prunes variants, retains caQTL/eQTL associations with both nominal $p > 0.05$, and examines correlations of their z-scores (Supplementary Figure S4).

TODO: Describe its descriptive role and distinguish SNP–pair observations from unique variants in the pooled summary.

Acknowledgments

TODO: Supply acknowledgments approved by the authors, if applicable. Keep funding and other submission-system declarations in their dedicated fields; see submission_requirements.md for the outstanding information.

References

1. Bowden J, Davey Smith G, Burgess S. Mendelian randomization with invalid instruments: effect estimation and bias detection through Egger regression. International Journal of Epidemiology.

2015;44(2):512-25. Available from: https://academic.oup.com/ije/article/44/2/512/754653. doi:10.1093/ije/dyv080.

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3. Liu Y, Xie J. Cauchy Combination Test: A Powerful Test With Analytic p-Value Calculation Under Arbitrary Dependency Structures. Journal of the American Statistical Association. 2020;115(529):393-402. Available from: https://doi.org/10.1080/01621459.2018.1554485. doi:10.1080/01621459.2018.1554485.

4. Benjamini Y, Hochberg Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. Journal of the Royal Statistical Society: Series B (Methodological). 1995;57(1):289-300. Available from: https://onlinelibrary.wiley.com/doi/10.1111/j.2517-6161.1995.tb02031.x. doi:10.1111/j.2517-6161.1995.tb02031.x.

5. Finucane HK, Bulik-Sullivan B, Gusev A, et al. Partitioning heritability by functional annotation using genome-wide association summary statistics. Nature Genetics. 2015;47(11):1228-35. Available from: https://www.nature.com/articles/ng.3404. doi:10.1038/ng.3404.

Supporting Information Captions

Supporting Table S1 is the copied regulatory-enrichment workbook `figures/Fig3_Hyper_table.xlsx`. Supporting Table S2 is the copied motif workbook `figures/Fig6_Motif_table.xlsx`. They remain native spreadsheets and are not rendered as LaTeX figures. Table S2 has updated numeric cells but inherited manual biological annotations; its "MR-only" title should be reviewed because the underlying foregrounds are method-specific.

TODO: Convert selected supporting-table content to manuscript tables after review; do not treat inherited annotation text as independently verified.

S1 Table. Regulatory-reference enrichment workbook.

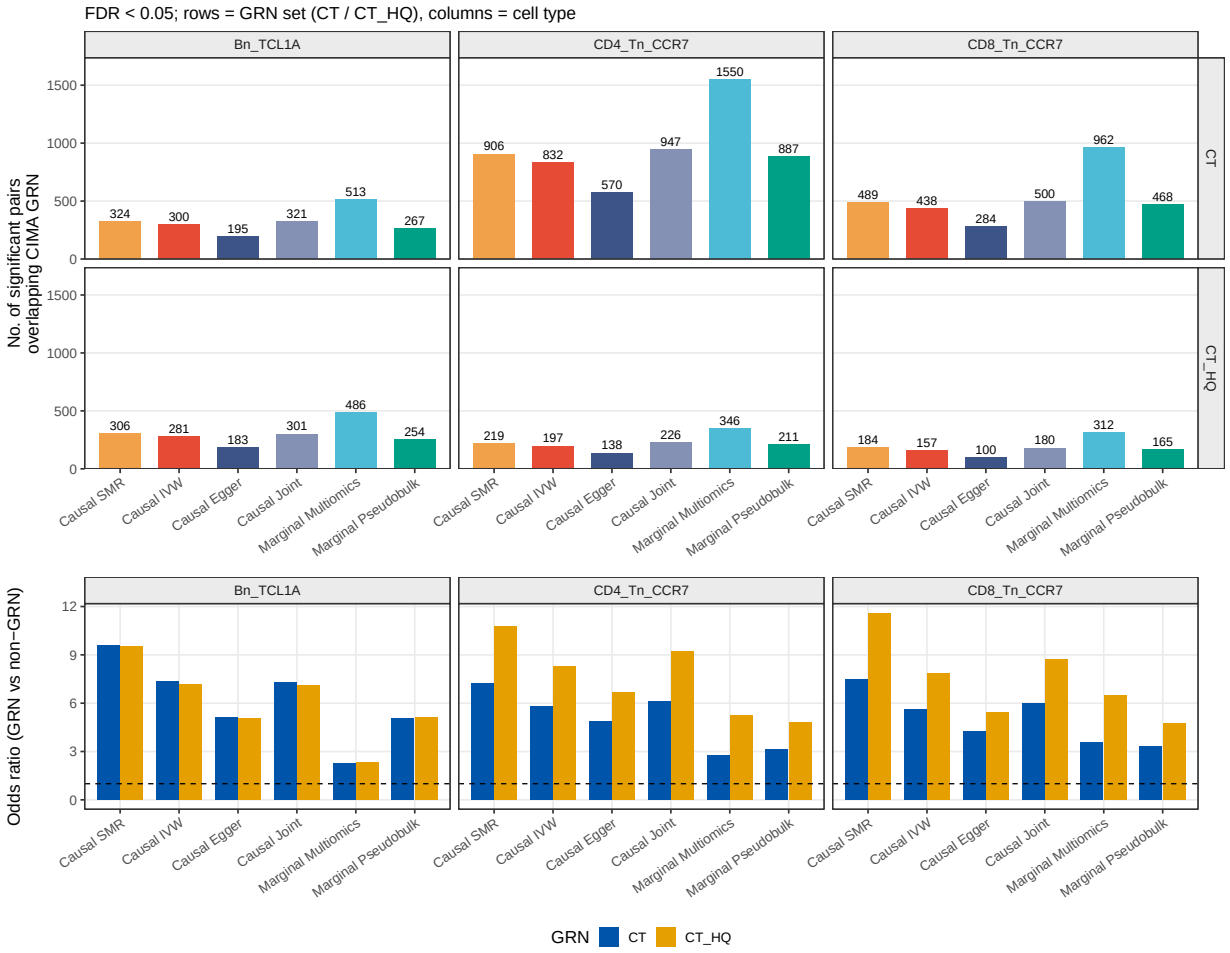
TODO: Finalize the supporting-table caption and file format after review of the existing workbook; no table content has been revised.

S2 Table. Motif enrichment workbook.

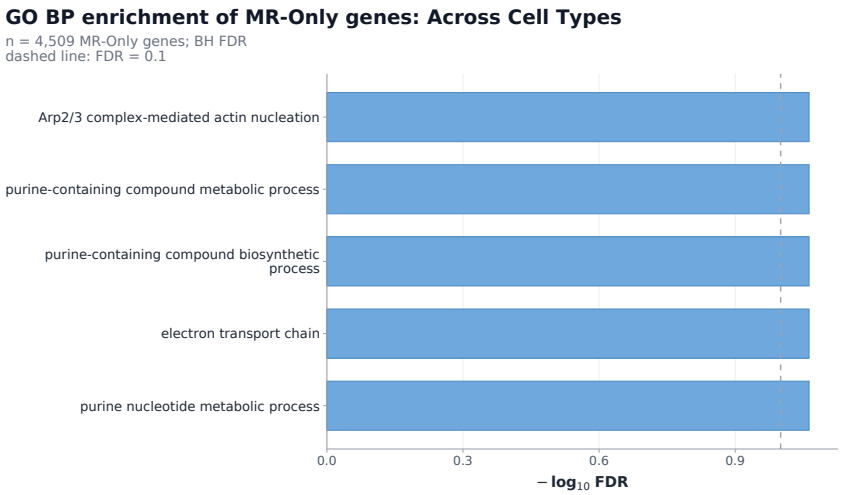
TODO: Finalize the supporting-table caption after review of the inherited annotations and foreground description noted above.

S1 Checklist. STROBE-MR reporting checklist (pending).

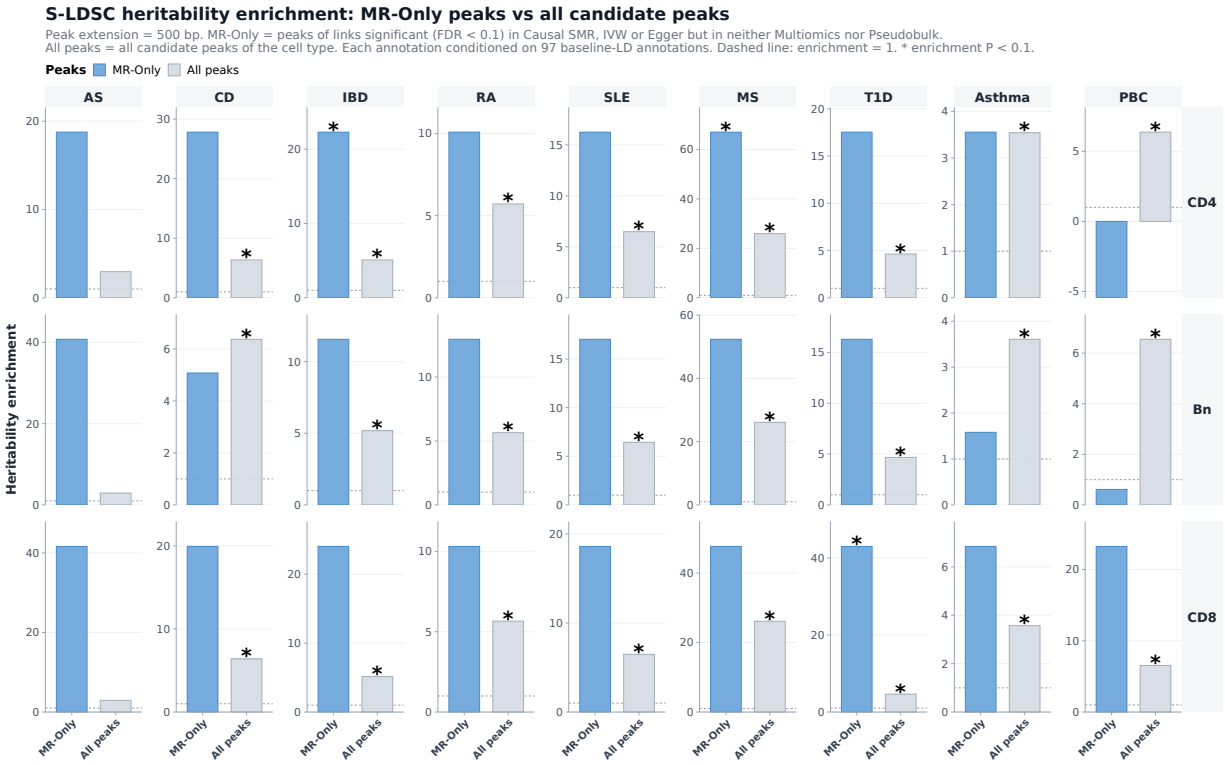
TODO: Prepare the completed reporting checklist with manuscript page numbers and relevant text. This label reserves a supporting item; no completed checklist is currently supplied.



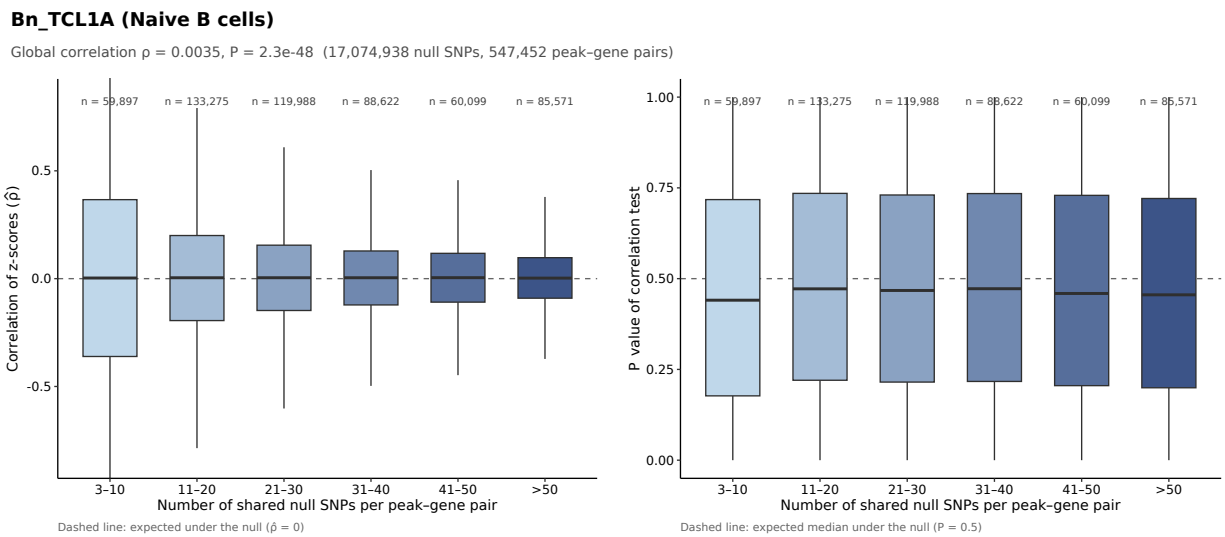
S1 Fig. Within-CIMA GRN corroboration. Top panels show significant-pair overlap counts at $q < 0.05$ for cell-type-active networks (CT) and their lineage high-quality subset (CT_HQ). Bottom panels show Fisher odds ratios relative to non-GRN tested pairs. This unchanged one-page figure is the latest available GRN artifact in the older figure directory.



S2 Fig. Conventional GO BP GSEA for the 4,509 combined MR-only target genes. The five terms with smallest conventional GSEA p-values are shown as $-\log_{10}$ BH-adjusted p-values; the dashed line marks adjusted $p = 0.1$. This is unchanged source page 2. Pages 1 and 3 of the copied PDF contain per-group conventional-p and empirical-p displays, respectively, and are not silently combined with this conventional comparison.



S3 Fig. Cell-type-specific S-LDSC enrichment. The unchanged source page 1 compares MROnly and all-candidate-peak annotations for CD4, Bn, and CD8, with the definition and raw-p asterisk convention of Figure 4. Axis scales differ between panels.



S4 Fig. Null-SNP diagnostic in Bn. Pair-level correlations of caQTL and eQTL z-scores, and their correlation-test p-values, are grouped by the number of shared pruned null SNPs. The displayed global summary pools SNP–pair observations; repeated variants may contribute to multiple pairs. This is unchanged source page 1, with CD4 and CD8 retained as pages 2 and 3 in the copied PDF. The diagnostic does not itself establish MR calibration.